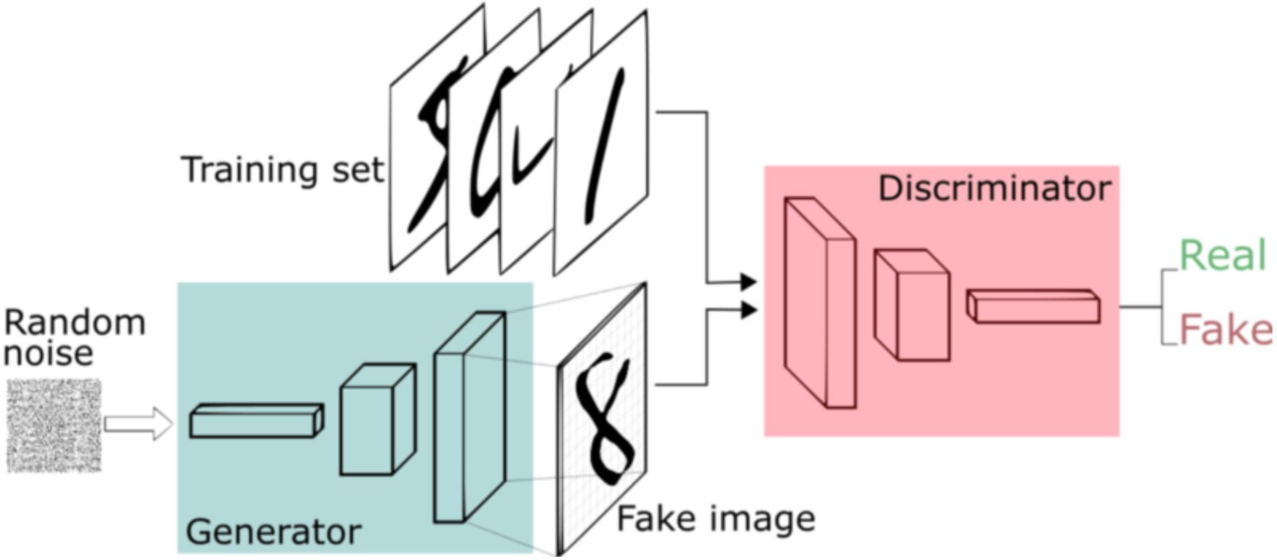
Getting started - GAN
(generative adversarial network)

GAN ideas

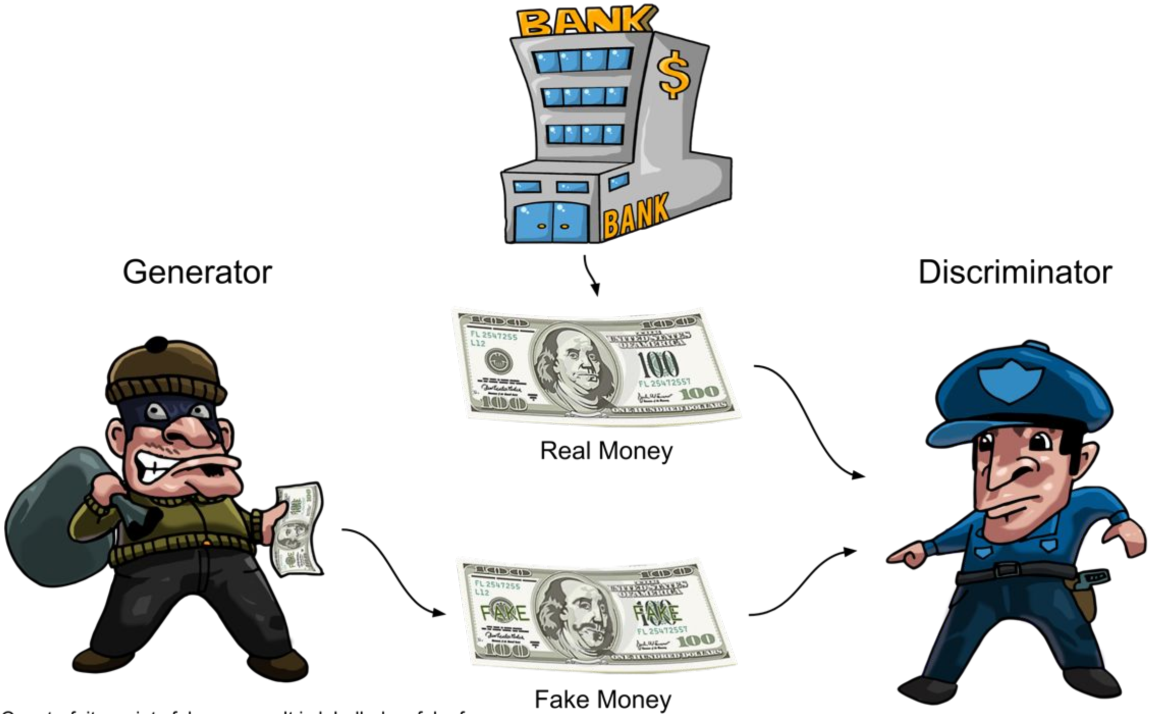
GAN Model

Generative Adversarial Networks



Generative Adversarial Network framework.

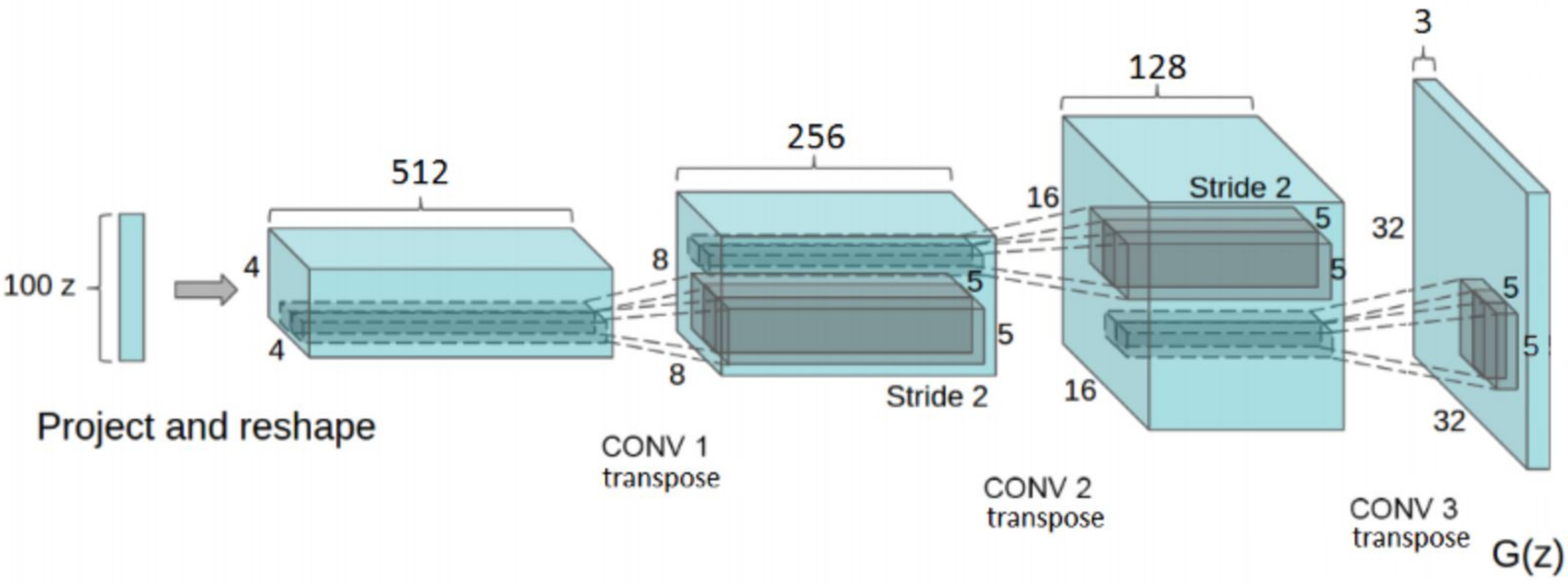
Generator vs Discriminator



Counterfeiter prints fake money. It is labelled as fake for police training. Sometimes, the counterfeiter attempts to fool the police by labelling the fake money as real.

The police are trained to spot real from fake money. Sometimes, the police give feedback to the counterfeiter why the money is fake.

Generator

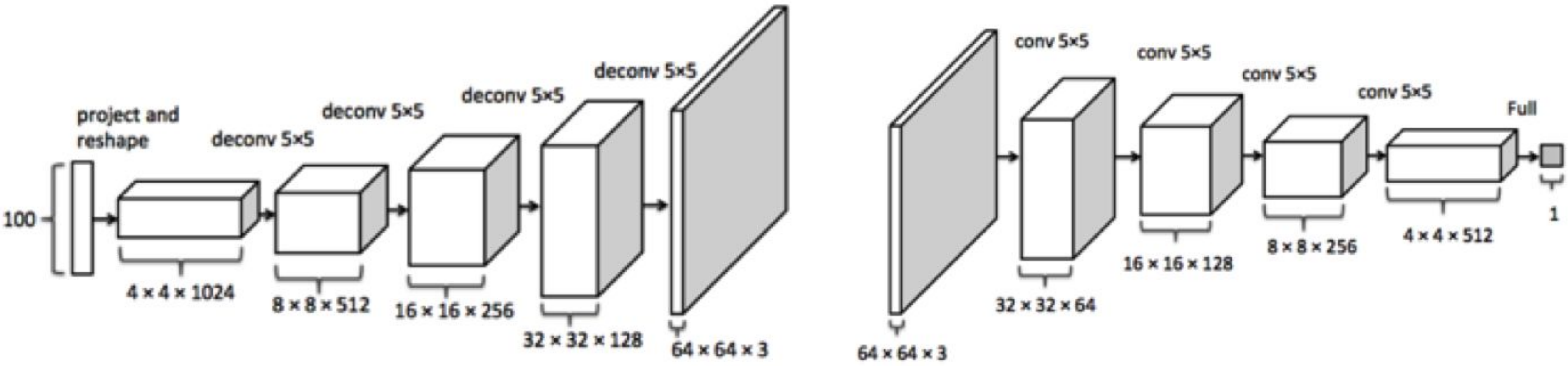


100 z
Project and reshape
512
4
4
256
8
8
5
5
Stride 2
CONV 1
transpose
128
16
16
Stride 2
5
5
CONV 2
transpose
3
32
32
5
5
CONV 3
transpose
G(z)

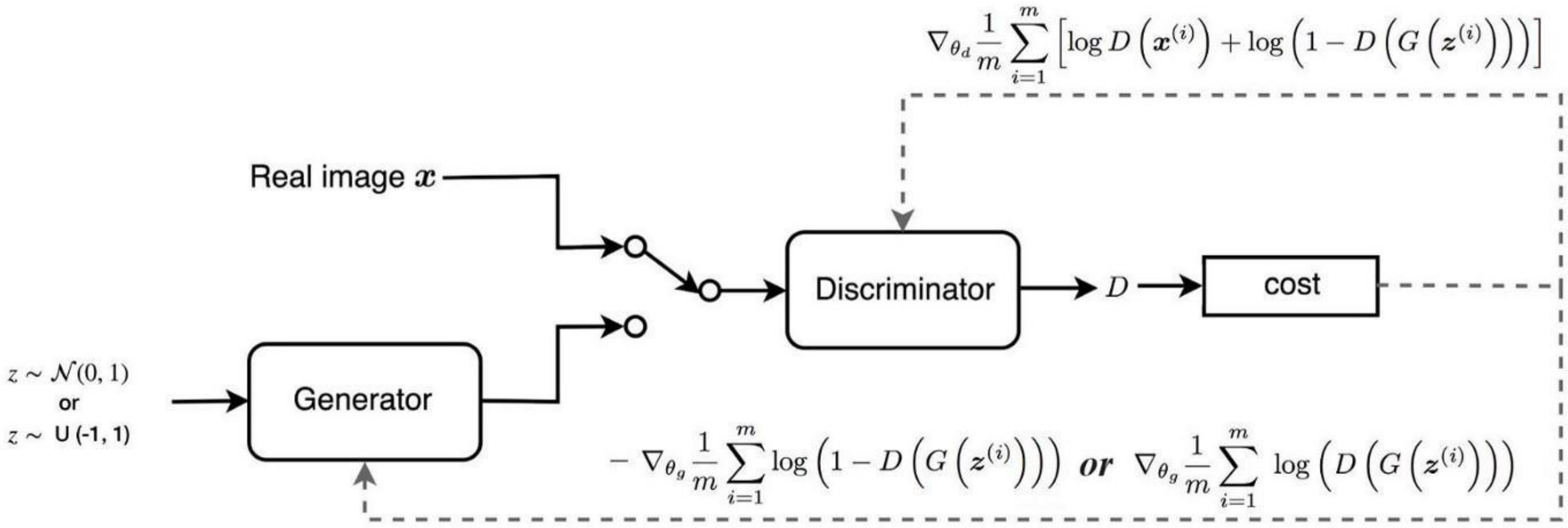
Full model

Generator

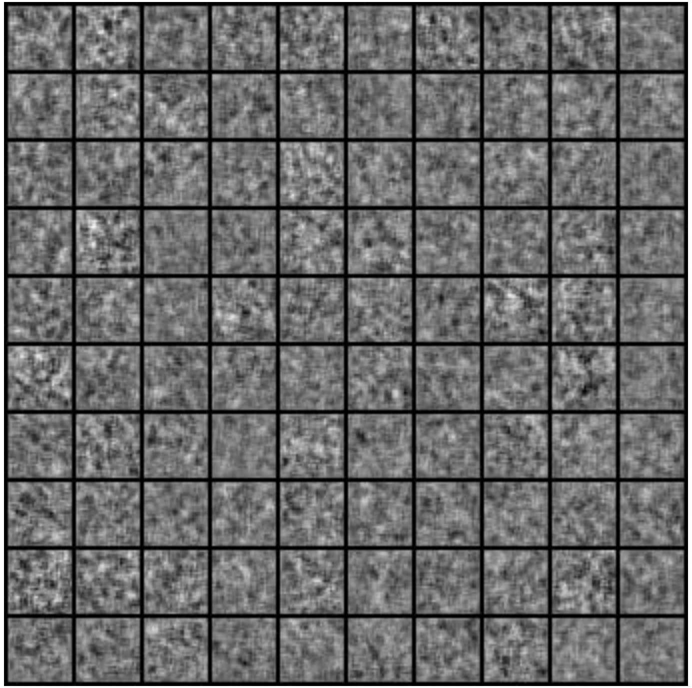
Discriminator



Loss function



Result



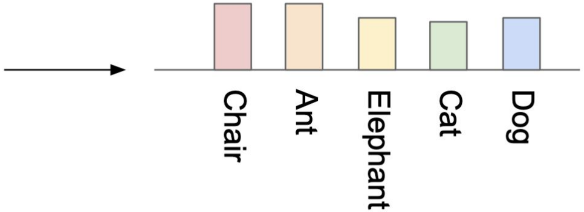
Evaluation

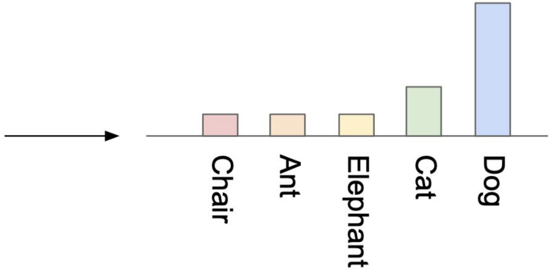
Diversity

100k steps

Quality

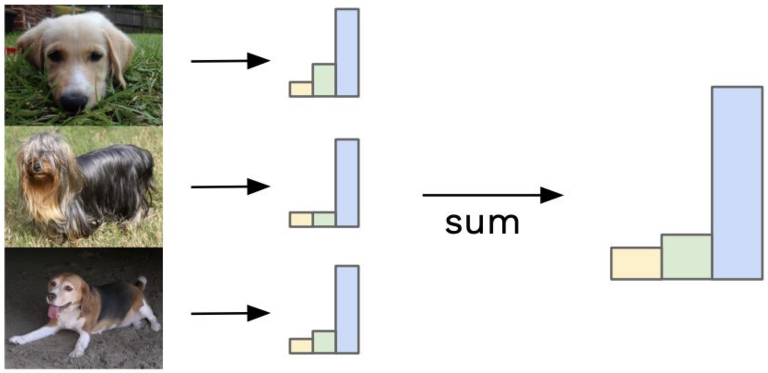
Inception Score (IS)



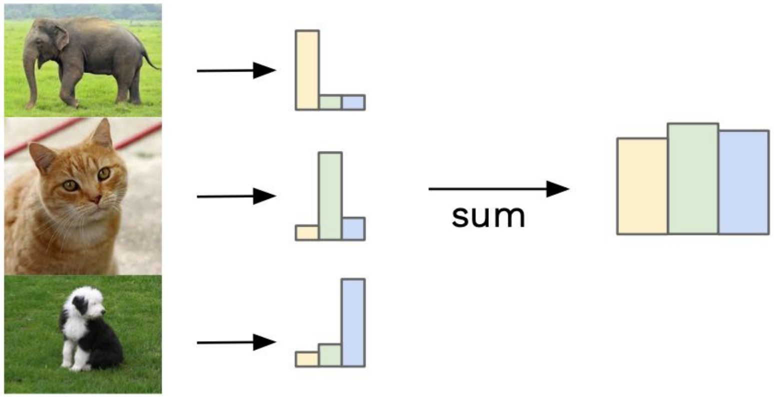


Inception Score (IS)

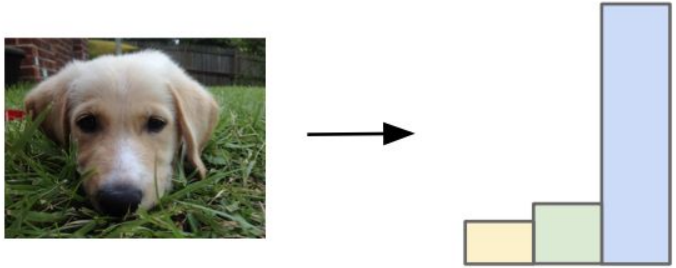
Similar labels sum to give focussed distribution



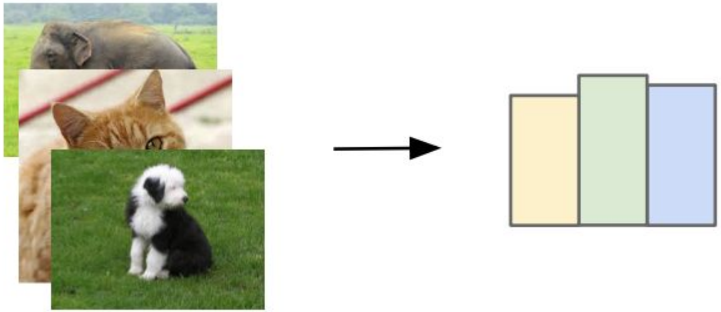
Different labels sum to give uniform distribution



Inception Score (IS)

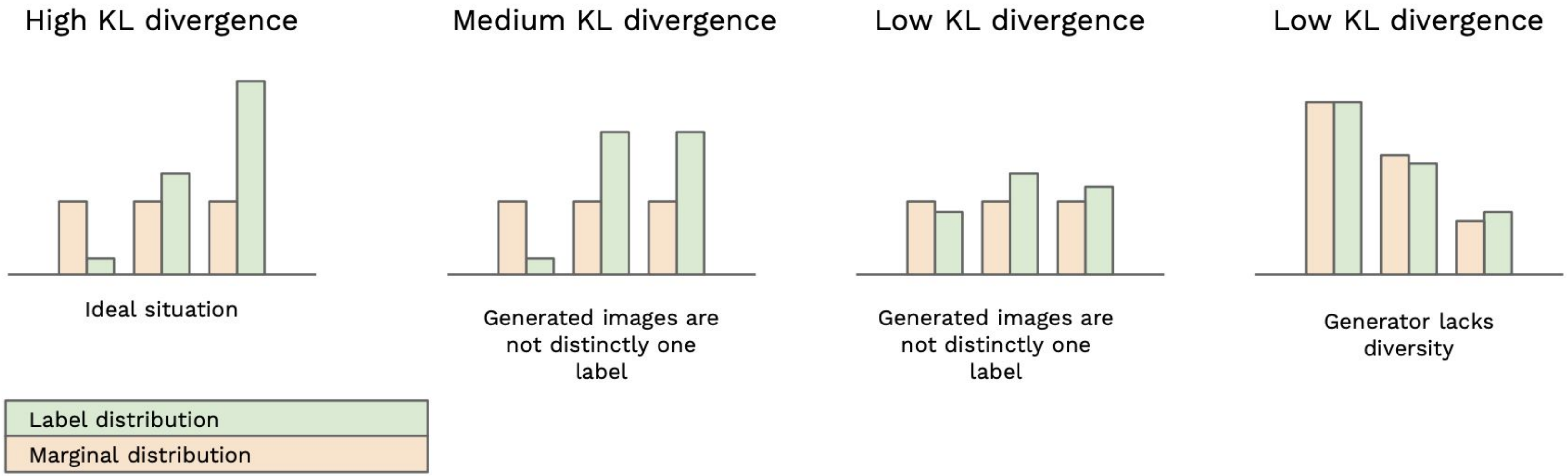


Ideal label distribution

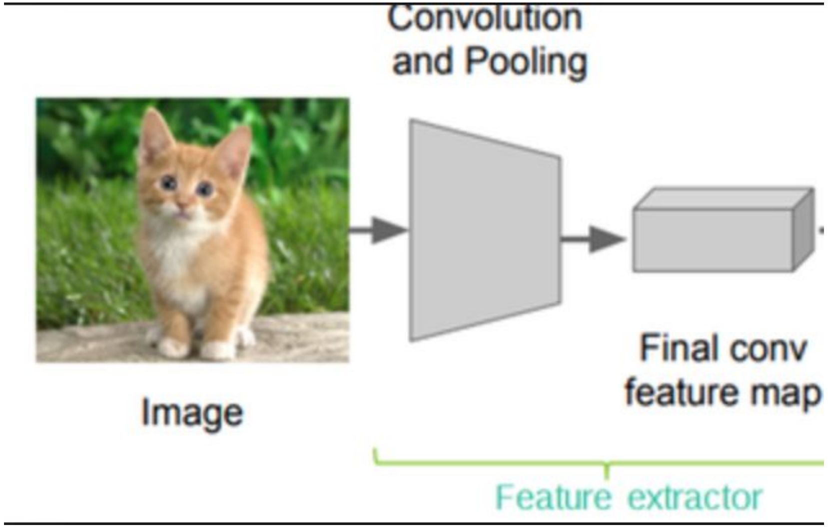


Ideal marginal distribution

Inception Score (IS)



Frechet Inception Distance (FID)



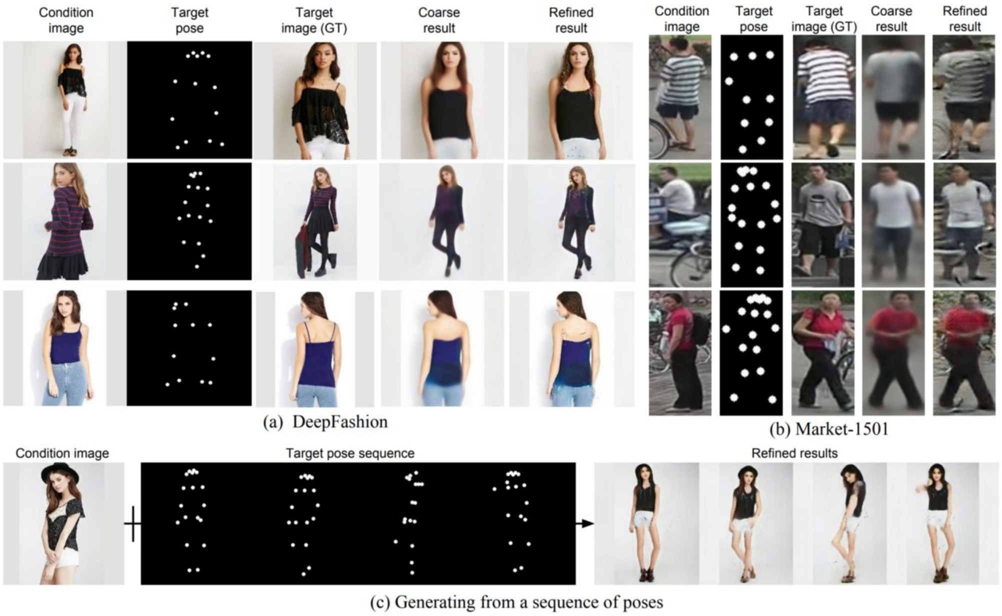
$$\texttt{FID}(x,g) = ||\mu_x - \mu_g||_2^2 + \mathrm{Tr}(\Sigma_x + \Sigma_g - 2(\Sigma_x\Sigma_g)^{\frac{1}{2}}),$$

Application

Anime character

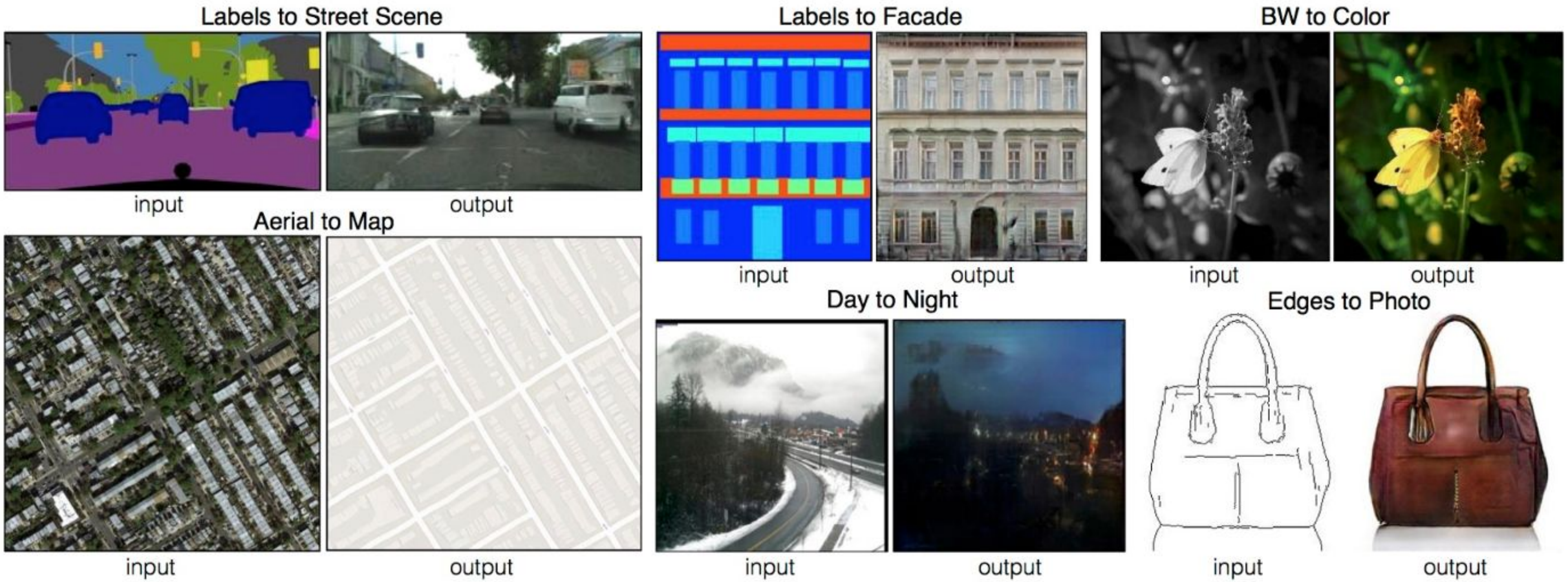
Figure 7: Generated samples

Pose guided person generation



Application

Pix2pix



Application

PGGAN

Figure 5: 1024 × 1024 images generated using the CELEBA-HQ dataset. See Appendix F for a larger set of results, and the accompanying video for latent space interpolations.

Stargan

Application

StackGAN

Application

CycleGAN

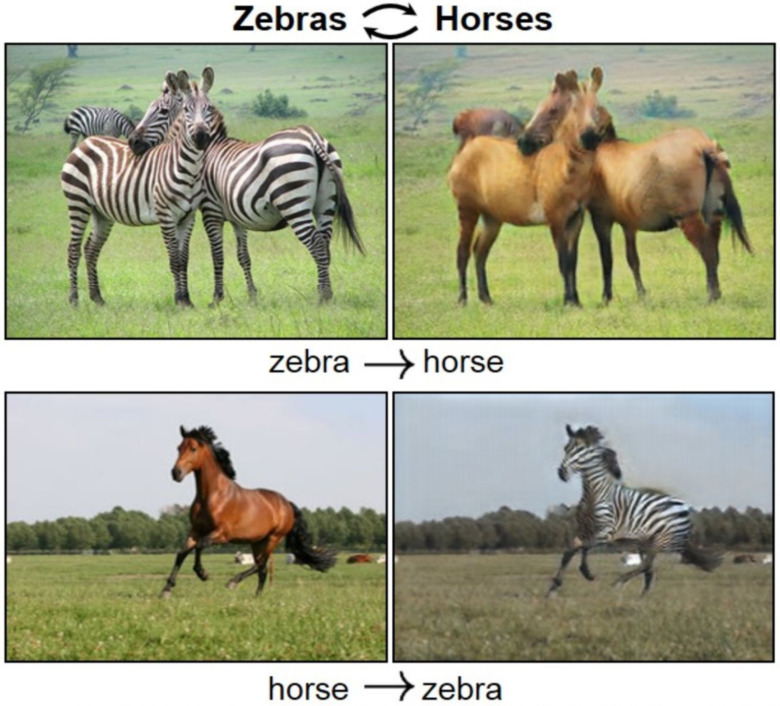
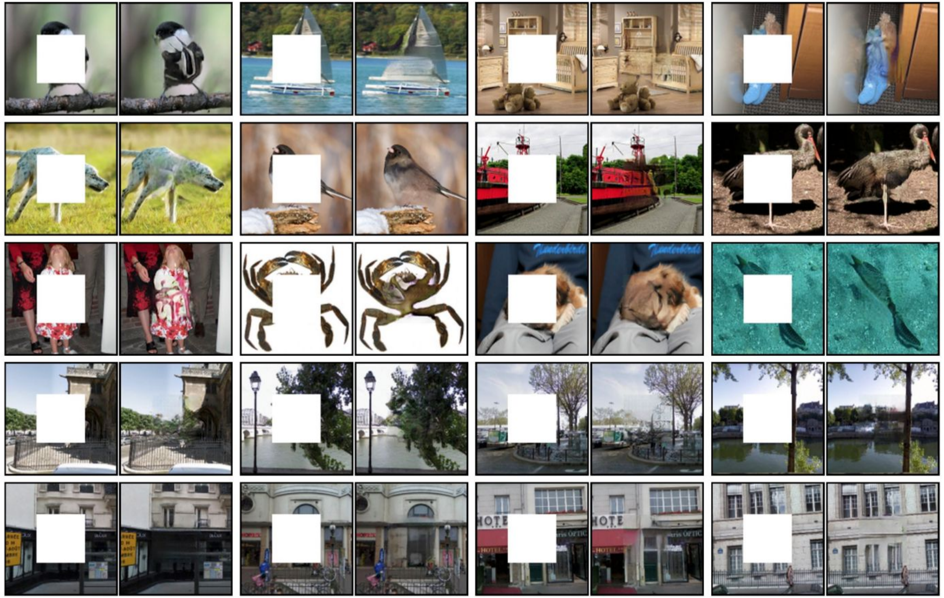


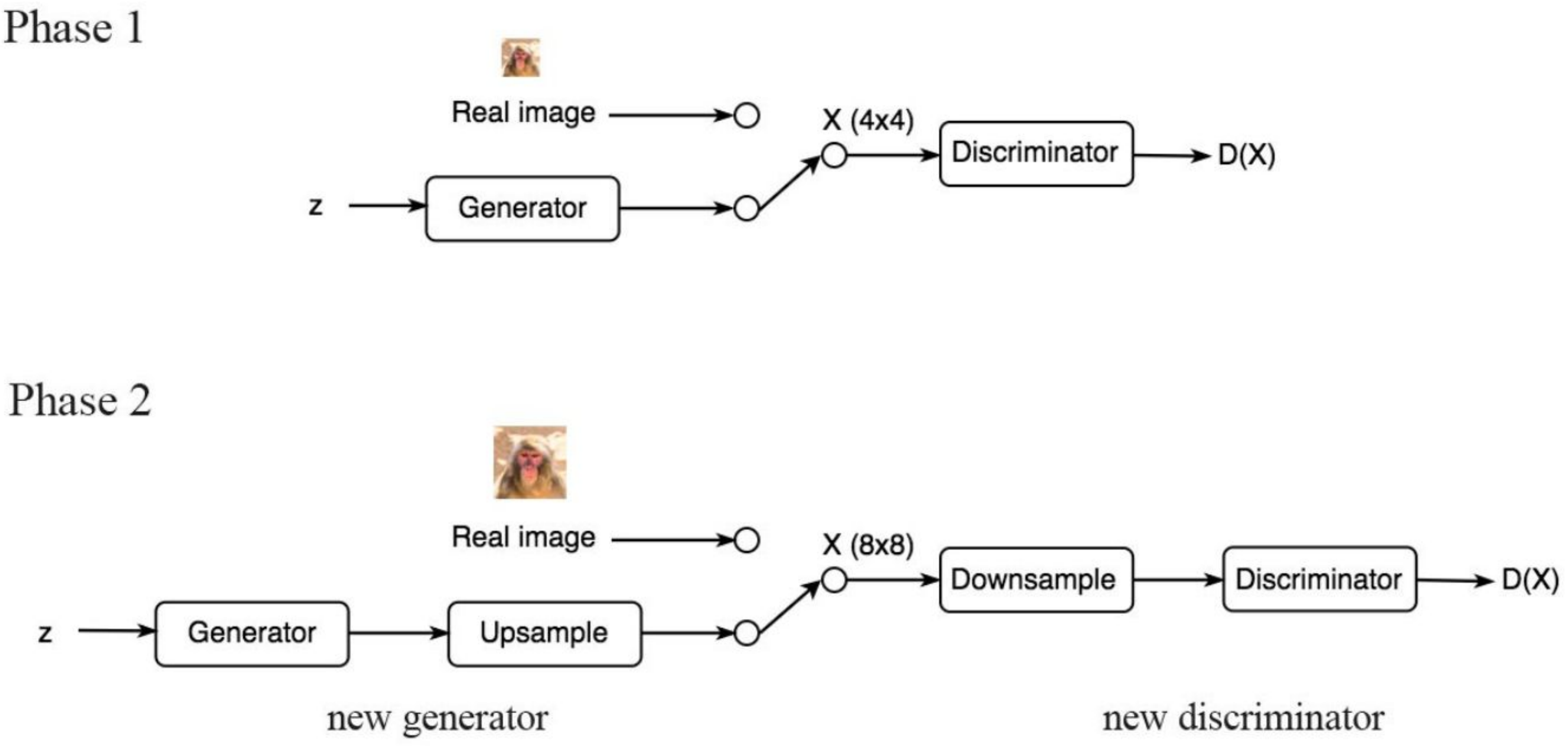
Image repainting



Progressive growing of GANs (PGGAN)

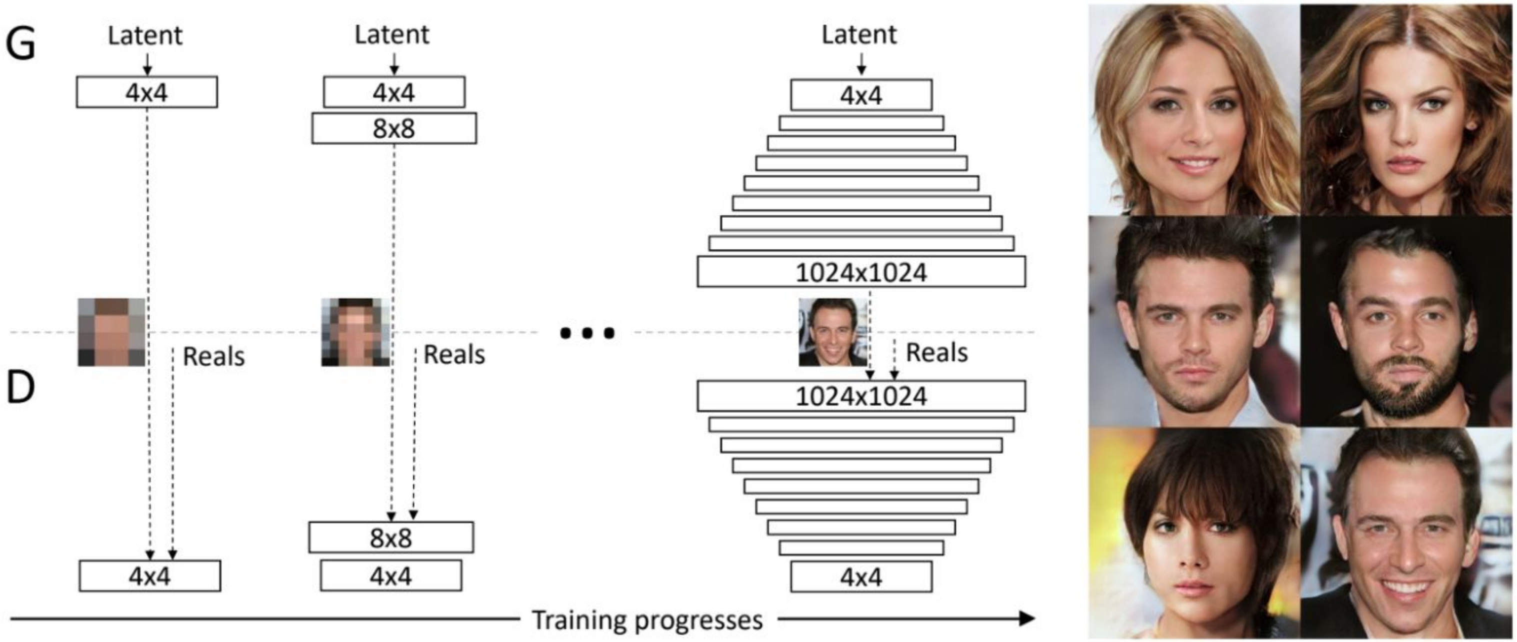
Figure 5: 1024×1024 images generated using the CELEBA-HQ dataset. See Appendix F for a larger set of results, and the accompanying video for latent space interpolations.

PGGAN

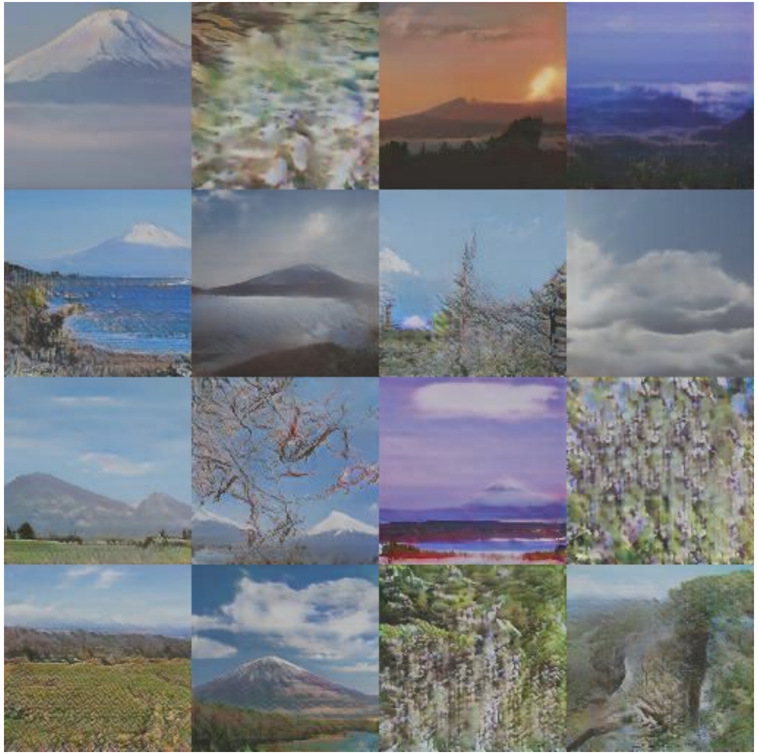


Phase 1
Real image
X (4x4)
Discriminator
D(X)
z
Generator
Phase 2
Real image
X (8x8)
Downsample
Discriminator
D(X)
z
Generator
Upsample
new generator
new discriminator

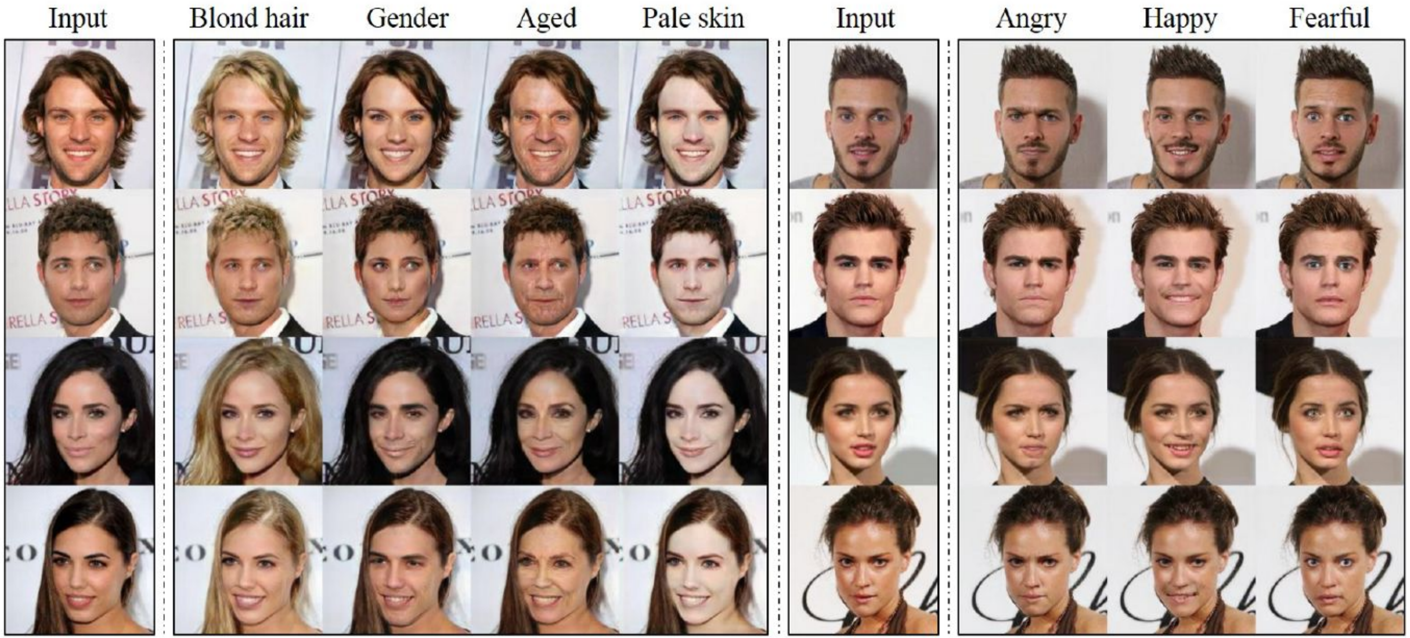
PGGAN


G
Latent
4x4
Latent
4x4
8x8
Latent
4x4
1024x1024
Reals
Reals
Reals
D
4x4
8x8
4x4
1024x1024
4x4
Training progresses

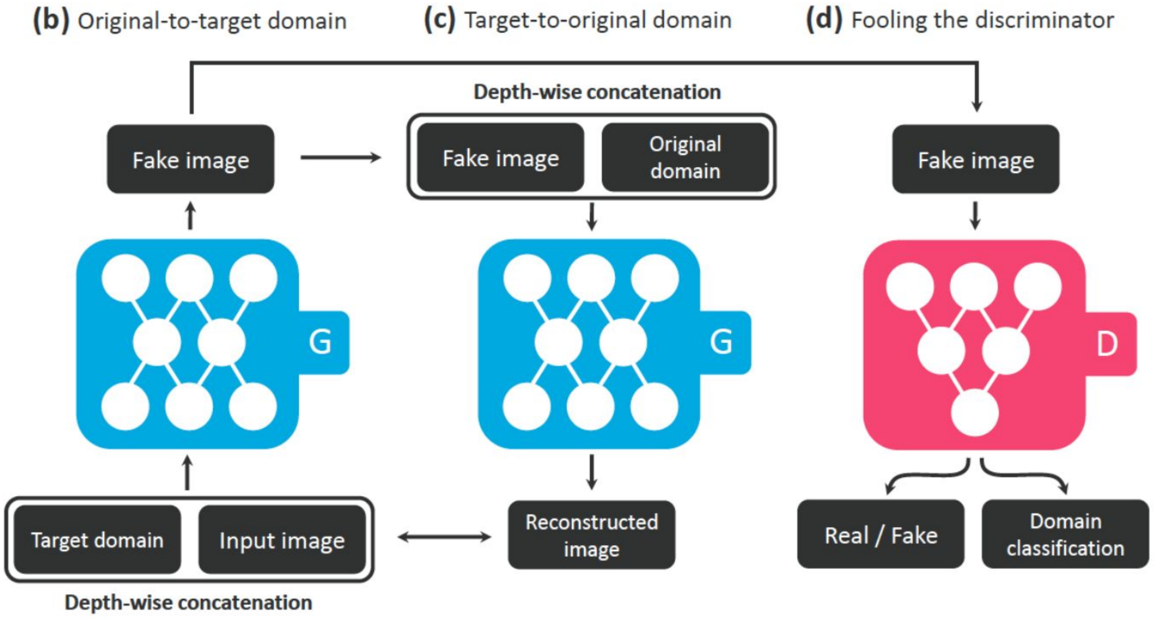
PGGAN



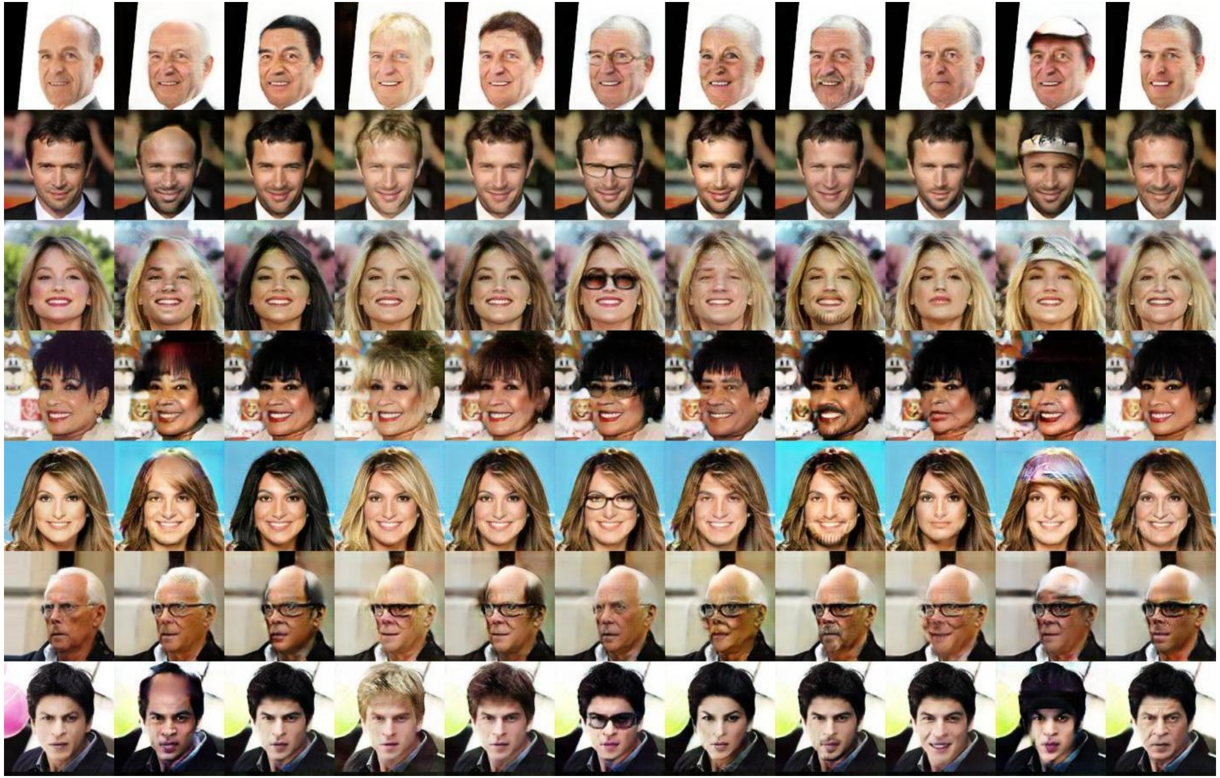
StarGAN


Input
Blond hair
Gender
Aged
Pale skin
Input
Angry
Happy
Fearful

StarGAN


(b) Original-to-target domain
(c) Target-to-original domain
(d) Fooling the discriminator
Depth-wise concatenation
Fake image
Fake image
Original domain
Fake image
G
G
D
Target domain
Input image
Reconstructed image
Real / Fake
Domain classification
Depth-wise concatenation

StarGAN




@LateNightSeth
THANK YOU!